Q1:

Adam has momentum terms beta1 and beta2.

First moment tracks gradients and second moment tracks variance.

Bias correction divides by beta1 to fix initial zero values.

SGD bounces around in ravines. Nesterov momentum helps prevent bouncing.

Adam changes learning rates automatically based on step size.

Q2:

~~Attention = softmax(Q * K / d_k)~~ * V.

Scaling by d_k makes numbers smaller so softmax does not explode.

Multihead attention splits vectors into multiple heads.

Complexity of attention is O(n * d). RNN complexity is O(n^2).

Q3:

Batch norm normalizes across samples. Layer norm normalizes layers.

Dropout drops nodes with probability p.

Inverted dropout multiplies by (1 - p).

Q4:

ResNet has shortcut connections: x + F(x).

Gradients can travel through shortcuts without shrinking.

Preactivation changes block ordering.